2NC---AT: PDB

1. PRESUMPTION FLIPS NEG---CP solves the case which means only a risk the perm is worse.

2. SAY NO---

a. QPQ---Delegation of authority to weaken IP protection is key.

Jose E. **Alvarez 21,** Professor of International Law, NYU School of Law. "Biden's International Law Restoration" International Law and Politics, Vol. 53:523. https://www.nyujilp.org/wp-content/uploads/2021/04/NYI205.pdf

Biden will not disrupt the United States' traditional reluctance to submit to supranational forms of adjudication. He is not likely to resolve the stalemate over the WTO's Appellate Body (AB) simply by agreeing to the appointment of new AB members and will insist on a quid pro quo for bringing the WTO's dispute settlement system back to life. Biden's nominee for U.S. Trade Representative (USTR), Katherine Tai, will probably not go as far as Trump's USTR, Robert Lighthizer, who proposed that the AB be abandoned in favor of singletiered arbitrations not correctable on appeal but subject to being overruled by WTO members in exceptional cases.102 But Tai, who helped to negotiate the USMCA, is likely to seriously consider proposals for more modest reforms. One set of proposals, backed by a group of states led by Ambassador David Walker of New Zealand and summarily rejected by Trump,103 suggests limiting the time for the AB to conclude appeals to ninety days, increasing the number of AB members to nine, requiring members to serve a single non-renewable term of eight years, restricting AB jurisdiction to issues that are necessary to resolve disputes, and establishing an annual meeting between WTO members and the AB to address systemic jurisprudential issues, such as whether AB rulings should be treated as de facto precedent.104

One of the most trenchant (albeit perhaps mostly symbolic) matters of WTO reform is getting China to agree that it should no longer be treated as a developing state. This issue might be subject to other tradeoffs that go through the WTO's dispute settlement system. For example, China might be persuaded to acquiesce to such a change if the United States accepts recent panel rulings like US-Tariff Measures on Certain Goods from China. 105 That ruling, which found U.S. tariff measures taken in response to intellectual property complaints directed at China to be illegal, echoes other recent decisions that require WTO states attempting to use the "essential security" exception of the General Agreement on Tariffs and Trade106 to do something more than invoke security as a magical dispensation from third-party scrutiny.107 It is possible that a more sober USTR might accept the old-fashioned idea that the United States gets reciprocal benefits when legally implausible arguments to defend protectionist actions fail to gain traction.108 Accepting such rulings might convince other WTO members that the United States still supports a rule-based system for trade and that its proposals for institutional reform are made in good faith.

b. HAND-TIPPING---secrecy is key to short and long-term trade negotiations.

Peter K. Yu 11, Kern Family Chair in Intellectual Property Law & Director, Intellectual Property Law Center, Drake University Law School; Wenlan Scholar Chair Professor, Zhongnan University of Economics and Law; Visiting Professor, Centre for International Intellectual Property Studies, University of Strasbourg. Six Secret (and Now Open) Fears of ACTA, 64 SMU L. Rev. 975 (2011). July. https://scholarship.law.tamu.edu/cgi/viewcontent.cgi?article=1437&context=facscholar //pipk

From a tactical standpoint, a secretive approach would make the negotiations smoother and much more successful. Such an approach would also help insulate the negotiations from external influences, which range from political complications in the capitols to opposition from civil society groups at both the national and international levels. In addition, by concealing the positions of each party, a secretive approach would greatly reduce the pressure faced by the negotiators while at the same time promoting an amicable long-term negotiating relationship. Such a relationship may benefit these negotiators in the long run, as it may go beyond the negotiation of this particular agreement or this particular version of the agreement. 192 For example, the benefits of an amicable relationship during the ACTA negotiations could be easily extended to cover the ongoing negotiation of the Trans-Pacific Partnership Agreement (TPP), which thus far have included among its negotiating parties Australia, Brunei Darussalam, Chile, Malaysia, New Zealand, Peru, Singapore, the United States, and Vietnam. 193

c. DEMOTIVATING---the plan means key groups won't lobby for the CP.

Aaditya Mattoo & Arvind Subramanian 9 Lead Economist in the Development Research Group of the World Bank. Senior Fellow at the Peterson Institute for International Economics and at the Center for Global Development. Jan/Feb. https://www.foreignaffairs.com/articles/2009-01-01/doha-next-bretton-woods //pipk

SINCE THE mid-1990s, world trade has grown rapidly, at a pace of approximately six percent a year — twice as fast as global economic output. During that time, however, WTO members have not adjusted the maximum levels of tariffs and other barriers that they can maintain on goods and services. In other words, overall trade has flourished, but the multilateral process that governs trade has languished.

Trade has grown throughout the world because many governments have increasingly come to believe that openness promotes long-term development. Many unilaterally liberalized their regulations on goods and services. Tariffs on goods have declined from a worldwide average of over 25 percent in 1980 to less than ten percent today. Many states have drastically reduced barriers to foreign investment and international trade in various service sectors, including finance, telecommunications, transport, and retail. Much of this liberalization has taken place in the context of regional trade agreements, such as the North American Free Trade Agreement (NAFTA) and a series of agreements between the European Union and its eastern neighbors. Since the early 1990s, the number of such pacts has risen from under 90 to nearly 400.

These parallel unilateral and regional efforts at liberalization ended up robbing the multilateral process of some of its raison d'être. By the time the Doha talks resumed in Geneva last summer, little of consequence was even on the table. In richer developing countries — a group that includes Brazil, China, and India — the reforms proposed would have left average tariff rates for agricultural goods unchanged, at about 13.5 percent, and would have reduced tariff rates for manufactured goods only slightly, from 6.4 percent to 5.6 percent.

Supporters of the Doha process concede that its goals when it comes to lowering trade barriers are modest but argue that its real purpose is to provide security for trading partners: legal commitments offer mutual assurances that trade policies will not be reversed. This argument would be compelling if the Doha talks actually contemplated guarantees against states suddenly resorting to punitively high import tariffs, such as those imposed by the Smoot-Hawley Tariff Act of 1930. In fact, the Doha proposals did not offer any meaningful guarantees of this kind. For example, under the proposals, the richer developing countries would still have had the leeway to adjust their agricultural tariffs by a margin of about 30 percentage points — and this is when their actual such tariffs are, on average, already 13.5 percent.

One sign of Doha's limited relevance is that the groups traditionally at the forefront of multilateral liberalization — private corporations in the intellectual-property, manufacturing, and service sectors — are now notable by their absence from the process. So modest were Doha's aims this past summer, that even the usual antiglobalization protesters did not bother to show up in Geneva.

3. INNOVATION DA---

a. the perm drains resources for innovation.

Peter K. Yu 11, Kern Family Chair in Intellectual Property Law & Director, Intellectual Property Law Center, Drake University Law School; Wenlan Scholar Chair Professor, Zhongnan University of Economics and Law; Visiting Professor, Centre for International Intellectual Property Studies, University of Strasbourg. Six Secret (and Now Open) Fears of ACTA, 64 SMU L. Rev. 975 (2011). July. https://scholarship.law.tamu.edu/cgi/viewcontent.cgi?article=1437&context=facscholar //pipk

Moreover, by creating a new forum for intellectual property enforcement, ACTA would take away the opportunity to strengthen enforcement norms in the current regimes-whether under the WTO or WIPO. In fact, the creation of multiple fora with overlapping jurisdictions has created serious challenges for national governments. As Kimberlee Weatherall reminded us: There is a risk of confusion and fragmentation in this process, particularly, one would think, for government departments and enforcement bodies subject to multiple overlapping requirements found in multiple overlapping agreements. In a context where we want government to be more efficient, subjecting them to multiple sources of regulation is not likely to lead to happy results. What would we rather government be doing-actually encouraging innovation, or box ticking on their customs processes to check compliance with the multiple different obligations in different treaties? What should money be spent on-grants for artists or yet forms and bodies and meetings about counterfeiting? 620 It is important to remember, given the limited resources, not every country will have the ability to undertake discussions in multiple fora. The more fora there are, the more difficult it will be for less-developed countries to dedicate their efforts to strengthening enforcement norms in a particular forum.

b. Innovation prevents extinction from climate change, that's Armstrong 21 from the 1NC.

4. BIDIRECTIONAL AUTHORITY KEY---authority to weaken protection is key to sustainable WTO adjudication

Peter K. Yu 11, Kern Family Chair in Intellectual Property Law & Director, Intellectual Property Law Center, Drake University Law School; Wenlan Scholar Chair Professor, Zhongnan University of Economics and Law; Visiting Professor, Centre for International Intellectual Property Studies, University of Strasbourg. Six Secret (and Now Open) Fears of ACTA, 64 SMU L. Rev. 975 (2011). July. https://scholarship.law.tamu.edu/cgi/viewcontent.cgi?article=1437&context=facscholar //pipk

More disturbing, the ACTA negotiators seemed to have assumed that the safeguards under existing U.S. law, such as fair use, the first sale doctrine, or the Bolar exception, will always stay in place under domestic law. Such assumptions, however, are questionable, especially in light of the increasing scrutiny of domestic legislation under the WTO process and the constantly evolving technological environment.

Consider, for example, the Fairness in Music Licensing Act, which expanded the unauthorized use of music by restaurants, bars, retail stores, and other small business establishments.50 6 Although that particular provision was enacted as a means to balance the extension of the copyright term in the Sonny Bono Copyright Term Extension Act,50 7 the WTO dispute settlement panel found the provision inconsistent with the United States' TRIPS obligations.508 Had the United States sought to comply with the panel decision by amending the Copyright Act-as compared to ignoring it509-the balance in the U.S. copyright system would tilt further toward rights holders.

Indeed, the WTO process has raised questions about the expediency and sustainability of the development of package legislation that includes strengthened protection and public interest offsets. As Graeme Dinwoodie and Rochelle Dreyfuss explained: Th[e] 'discrete' approach to adjudication (by which we mean that discrete parts of legislative compromises are broken out for individual assessment) can produce perverse consequences. Not only does it unravel carefully negotiated legislative deals, it does so in a systematic way. Because TRIPS sets only minimum standards, WTO dispute resolution operates as a one-way ratchet: complaints can lead to the invalidation of measures that reduce the level of intellectual property protection, but they never reach measures that increase protection. Thus, compromises will always unravel in the same direction, requiring nations to change those features of their legislation that benefit user groups while protection-enhancing provisions stay in place.5 10 Thus, according to Professors Dinwoodie and Dreyfuss, the discrete approach taken by the WTO dispute settlement panels would have the perverse effect of encouraging intellectual property rights holders and their supportive governments "to agree to provisions that reduce the level of protection in exchange for the protection-enhancing legislation that they want, knowing that the reductions will be successfully challenged at the international level."'511

5. FORCED COMPLIANCE KEY---The perm doesn't establish WTO rulings as self-executing, prevents compliance with future rulings

Peter K. Yu 11, Kern Family Chair in Intellectual Property Law & Director, Intellectual Property Law Center, Drake University Law School; Wenlan Scholar Chair Professor, Zhongnan University of Economics and Law; Visiting Professor, Centre for International Intellectual Property Studies, University of Strasbourg. Six Secret (and Now Open) Fears of ACTA, 64 SMU L. Rev. 975 (2011). July. https://scholarship.law.tamu.edu/cgi/viewcontent.cgi?article=1437&context=facscholar //pipk

Third-and the reason why many people hate lawyers-conflicts between the agreement and domestic law will not arise based on how laws operate in the participating countries. For example, in a so-called dualist jurisdiction like the United States, international agreements are rarely self-executing-that is, treaties can rarely be enforced in courts without the support of prior implementing legislation from Congress.268 Because ACTA was negotiated as a sole executive agreement, it virtually guarantees that it will not modify any domestic law.269 As the USTR stated in its fact sheet in the early days of the negotiations, "[o]nly the U.S. Congress can change U.S. law. ' '270 That statement, of course, is not entirely correct in a common law jurisdiction where judges do make law. However, it does state the correct legal conclusion that a sole executive agreement cannot change U.S. law.

In this scenario, if conflicts exist between the agreement and domestic law, domestic law will prevail and will remain intact until Congress chooses to amend the law. In fact, the United States can always choose not to fully implement an international agreement, its reputation and standing in the international community notwithstanding. For example, section 110(5)(B) of the U.S. Copyright Act remains intact even though a WTO dispute settlement panel has found the Fairness in Music Licensing Act inconsistent with the TRIPS Agreement.2 71 Likewise, despite an adverse WTO panel decision, the United States has yet to amend section 211(a)(2) of the U.S. Omnibus Appropriations Act of 1998, which prohibits the registration or renewal of trademarks previously abandoned by trademark holders whose business and assets have been confiscated under Cuban law. 272 In both situations, Congress refused to change its laws even though the United States is in breach of WTO rules. 273 Obviously, if the United States chooses not to honor its commitments, there is an additional question concerning what other countries could do under ACTA. Unlike the WTO, ACTA does not possess any enforcement mechanism. Instead, Article 38 merely outlines the consultation process between the signatory parties over matters affecting the implementation of the agreement. 274 Thus, structurally, the agreement is closer to an organized, treaty-based dialogue than a plurilateral trade agreement (although the U.S. administration thus far has remained reluctant to admit ACTA's very limited legal effect on U.S. law). 275

Finally-and the reason why people hate politicians-the negotiators might have taken the oft-used diplomatic position that "nothing is agreed until everything is agreed. '2 76 In short, the negotiators might have had a good-faith belief that they could succeed in convincing their negotiating partners to remove those brackets that required modification of domestic law in their specific country. Such beliefs were not entirely unfounded from negotiators from countries with very strong bargaining power, such as the United States and members of the European Union. Nevertheless, the strong disagreement between these two trading powers over various intellectual property areas has raised serious competence issues. Indeed, many of these disagreements-such as the scope of coverage of the section on border measures-last until the very end.277

2NC---AT: PDCP

1. CP isn't the plan.

a. We ban strengthening federal protection of domestic rights as a violation of maximum standards under TRIPS. That's key to the net benefit. Cross-apply the perception link from above, that's Yu 11.

b. Any counterplan could result in solving the aff. If they win the CP isn't competitive because the aff doesn't have to be certain, kick the CP and vote neg on presumption because if they don't fiat certainty, then inherency means it won't happen.

2. Severance is a voting issue. It's a no-cost option that incentivizes spamming terrible, poorly worded perms hoping we drop one that they get to re-explain in the 1AR. Bad for neg ground because it discourages counterplans, bad for clash because it discourages aff research, bad for fairness because it moots the block as a structural response to aff side biases.

3. CP doesn't strengthen its protection

a. "It's" means belonging to

Merriam-Webster's Learner's Dictionary, No Date, http://www.merriam-webster.com/dictionary/its

its

adjective \ˈits, əts\

Simple Definition of its

: relating to or belonging to a certain thing, animal, etc. : made or done by a certain thing, animal, etc.

b. And, is exclusive

Brent 10 (Douglas F. Brent, attorney, "Reply Brief on Threshold Issues of Cricket Communications, Inc.," 6-2-2010, http://psc.ky.gov/PSCSCF/2010%20cases/2010-00131/20100602_Crickets_Reply_Brief_on_Threshold_Issues.pdf)

AT&T also argues that Merger Commitment 7.4 only permits extension of "any given" interconnection agreement for a single three year term. AT&T Brief at 12. Specifically, AT&T asserts that because Cricket adopted the interconnection agreement between Sprint and AT&T, which itself was extended, Cricket is precluded from extending the term of its agreement with AT&T. Id This argument relies upon an inaccurate assumption: that the agreement (contract) between Sprint and AT&T, and the agreement (contract) between Cricket and AT&T, are one and the same. In other words, to accept AT&T's argument the Commission must conclude that two separate contracts, i.e. the interconnection between Sprint and AT&T in Kentucky ("Sprint Kentucky Agreement") and the interconnection between Cricket and AT&T in Kentucky ("Cricket Kentucky Agreement"), are one and the same. Upon this unstated (and inaccurate) premise

AT&T asserts that “the ICA was already extended”; id. at 14, and “the ICA Cricket seeks to extend was extended by Sprint . . . .”; id. at 15, and, finally, “Cricket cannot extend the same ICA a second time . . . .” Id. (emphasis added in all). Note that in the quoted portions of the AT&T brief (and elsewhere) **AT&T uses vague and imprecise language** when referring to either the Sprint Kentucky Agreement, or the Cricket Kentucky Agreement, **in hopes that the Commission will treat the two contracts as one and the same**. But it would be a mistake to do so. The contract governing AT&T’s duties and obligations with Sprint is a legally distinct and separate contract from that which governs AT&T’s duties with Cricket. The Sprint Kentucky Agreement was approved by the Commission in September of 2001 in Case Number 2000-00480. The Cricket Kentucky Agreement was approved by the Commission in September of 2008 in Case Number 2008-033 1. **AT&T ignores the fact that these are two separate and distinct contracts because it knows that the merger commitments apply to each agreement** that an individual telecommunications carrier has with AT&T. Notably, Merger Commitment 7.4 states that “AT/BellSouth ILECs shall permit a requesting telecommunications carrier to extend its current interconnection agreement . . . . As written, **the commitment allows any carrier to extend “its” agreement**. Clearly, **the use of the pronoun “its” in this context is possessive, such that the term “its” means - that particular carrier’s agreement** with AT&T (**and not any other carrier’s agreement**). Thus, the merger commitment applies to each agreement that an individual carrier may have with AT&T. It necessarily follows then, that Cricket’s right to extend its agreement under Merger Commitment 7.4 is separate and distinct right from another carrier’s right to extend its agreement with AT&T (or whether such agreement has been extended).

4. CP isn’t domestic

Peter **Gerhart 4**, Professor of Law, Case Western Reserve. SYMPOSIUM: "THE FUTURE OF INTERNATIONAL INTELLECTUAL PROPERTY: THE INTERNATIONAL RELATIONS OF INTELLECTUAL PROPERTY LAW": INTRODUCTION: THE TRIANGULATION OF INTERNATIONAL INTELLECTUAL PROPERTY LAW: COOPERATION, POWER, AND NORMATIVE WELFARE, 36 Case W. Res. J. Int'l L. 1 //pipk

By and large, the core concern of intellectual property scholars who study international intellectual property is normative. They want to know how we can permit the broadest possible access to existing knowledge without undercutting the incentives needed to generate a high rate of new knowledge. This substantive, normative focus on the appropriate balance between the incentive effects and the access effects of intellectual property [*4] is the central concern of domestic intellectual property -- and heretofore the concern of states acting in relative isolation. Now that markets, including markets for knowledge goods, are transnational and global, that traditional question is transposed to the family of states. Here, the principle of territoriality -- that is, the principle that intellectual property law is primarily the creation of states and is confined to the borders of states -- induces intellectual property scholars to ask the following central question: is the sum of the intellectual property systems of heterogeneous states, as influenced through international intermediation and institutions, enhancing global welfare or not? The focus is inevitably normative, with some metric of social welfare serving as the focal point for analysis. 9

2NC---!---Growth

Turns growth. Trade is also key to global productivity, preventing a decades long economic slowdown.

World Bank 18 – World Bank, Press Release 1-9-2018, "Global Economy to Edge Up to 3.1 percent in 2018 but Future Potential Growth a Concern ," https://www.worldbank.org/en/news/press-release/2018/01/09/global-economy-to-edge-up-to-3-1-percent-in-2018-but-future-potential-growth-a-concern)

WASHINGTON, January 9, 2018— The World Bank forecasts global economic growth to edge up to 3.1 percent in 2018 after a much stronger-than-expected 2017, as the recovery in investment, manufacturing, and trade continues, and as commodity-exporting developing economies benefit from firming commodity prices.

However, this is largely seen as a short-term upswing. Over the longer term, slowing potential growth—a measure of how fast an economy can expand when labor and capital are fully employed—puts at risk gains in improving living standards and reducing poverty around the world, the World Bank warns in its January 2018 Global Economic Prospects.

Growth in advanced economies is expected to moderate slightly to 2.2 percent in 2018, as central banks gradually remove their post-crisis accommodation and as an upturn in investment levels off. Growth in emerging market and developing economies as a whole is projected to strengthen to 4.5 percent in 2018, as activity in commodity exporters continues to recover.

"The broad-based recovery in global growth is encouraging, but this is no time for complacency," World Bank Group President Jim Yong Kim said. "This is a great opportunity to invest in human and physical capital. If policy makers around the world focus on these key investments, they can increase their countries' productivity, boost workforce participation, and move closer to the goals of ending extreme poverty and boosting shared prosperity."

Download the January 2018 Global Economic Prospects report.

2018 is on track to be the first year since the financial crisis that the global economy will be operating at or near full capacity. With slack in the economy expected to dissipate, policymakers will need to look beyond monetary and fiscal policy tools to stimulate short-term growth and consider initiatives more likely to boost long-term potential.

The slowdown in potential growth is the result of years of softening productivity growth, weak investment, and the aging of the global labor force. The deceleration is widespread, affecting economies that account for more than 65 percent of global GDP. Without efforts to revitalize potential growth, the decline may extend into the next decade, and could slow average global growth by a quarter percentage point and average growth in emerging market and developing economies by half a percentage point over that period.

"An analysis of the drivers of the slowdown in potential growth underscores the point that we are not helpless in the face of it," said World Bank Senior Director for Development Economics, Shantayanan Devarajan. "Reforms that promote quality education and health, as well as improve infrastructure services could substantially bolster potential growth, especially among emerging market and developing economies. Yet, some of these reforms will be resisted by politically powerful groups, which is why making this information about their development benefits transparent and publicly available is so important."

Risks to the outlook remain tilted to the downside. An abrupt tightening of global financing conditions could derail the expansion. Escalating trade restrictions and rising geopolitical tensions could dampen confidence and activity. On the other hand, stronger-than-anticipated growth could also materialize in several large economies, further extending the global upturn.

“With unemployment rates returning to pre-crisis levels and the economic picture brighter in advanced economies and the developing world alike, policymakers will need to consider new approaches to sustain the growth momentum,” said World Bank Development Economics Prospects Director Ayhan Kose. “Specifically, productivity-enhancing reforms have become urgent as the pressures on potential growth from aging populations intensify.”

In addition to exploring developments at the global and regional levels, the January 2018 Global Economic Prospects takes a close look at the outlook for potential growth in each of the six global regions; lessons from the 2014-2016 oil price collapse; and the connection between higher levels of skill and education and lower levels of inequality in emerging market and developing economies.

Judiciary ADV

1 Court Stability—2NC/1NR

A) ***Alt Cause — judge shopping is rampant. Litigators can get cases in front of sympathetic judges which erodes court credibility and stability by ruining public trust. That’s Kanu.***

Political affiliations of federal circuit judges erode court stability.

Cohen 24 — Alma Cohen, Professor of Empirical Practice at Harvard Law School, Professor at The Eitan Berglas School of Economics at Tel Aviv University, Faculty Research Fellow at the National Bureau of Economic Research, holds a Ph.D. in Economics from Harvard University, holds an M.A. in Economics from Harvard University, holds a B.A. in Mathematics from Tel Aviv University, holds an M.Sc. in Management and Operations Research from Tel Aviv University, 2024 (“The pervasive influence of political composition on circuit court decisions,” *Center for Economic and Policy Research*, February 13th, Available Online at https://cepr.org/voxeu/columns/pervasive-influence-political-composition-circuit-court-decisions, Accessed 09-08-2024)

It is widely believed that the identity of the US presidents appointing justices to the country’s Supreme Court can help to predict the justices’ decisions in the small number of ideologically salient issues the court considers each year. But what about the country’s federal Circuit Courts of Appeals? These courts, which are one level below the Supreme Court, consider all appeals on the rulings of federal district courts; each year they decide many thousands of cases that mostly do not seem to involve ideologically controversial cases. When, and to what extent, do the decisions of circuit court judges that are appointed by Democratic presidents (henceforth, ‘Democratic judges’) and Republican presidents

('Republican judges') systematically differ? This question has been the subject of a long-standing and heated debate.

The 'traditional' view is probably best represented by a series of articles written long ago by two former chief judges of the Circuit Court of Washington, DC (e.g. Edwards 1985, Wald 1994). These judges conceded that **political affiliations are often associated with judicial decisions** at the Supreme Court, but they maintained that this was largely not the case at the circuit courts. In their view, the political affiliations of circuit court judges are irrelevant to the outcomes of most circuit court cases.

By contrast, a 'legal realist' view on the subject has been developed, based partly on empirical research, by a number of prominent legal and political science scholars. Classic studies documented the existence of 'party effects' in a set of cases involving free speech, civil rights, labour relations, and criminal appeals (Songer and Davis 1990); in a set of cases reviewing decisions by the Environmental Protection Agency (Revesz 1997); and in sets of cases on 'ideologically controversial' issues, such as abortion, affirmative action, capital punishment, and sex discrimination (Sunstein et al. 2004). Subsequent work confirmed this pattern of party effects in various categories of cases that are ideologically controversial or salient. Although these scholars argued that there is evidence that the **political composition of circuit court panels can predict outcomes** in some specific sets of cases, they mostly left unanswered the question of how broad and widespread the influence of political composition is. For example, Sunstein et al. (2004) stressed that their findings were "limited to domains where ideology would be expected to play a larger role", and that "outside of such domains, Republican and Democratic appointees are far less likely to differ". These authors remained agnostic on whether party effects might be present in "apparently non-ideological cases involving, for example, bankruptcy, torts, and civil procedure", and they viewed answering this question as an important challenge for future work

In a recent study (Alma 2023), I seek to meet this challenge. My ability to do so is facilitated by a novel dataset I have compiled, which is much larger and more comprehensive than the datasets used by earlier works on the subject. Prior empirical works have largely focused on limited sets of cases that involved ideologically salient issues and that had a published opinion. By contrast, my dataset includes about 670,000 cases from the period 1985-2020. This dataset encompasses all the varied types of cases that are considered by the circuit courts, including cases that are not ideologically salient and cases without a published opinion.

Using this dataset, I investigate whether significant party effects are present in the vast universe of circuit court cases. There are several reasons to expect Democratic and Republican judges to systematically differ in such cases. Among other things, cases that are not ideologically salient might still involve some ideological dimensions; Democratic and Republican judges could systematically differ in their attitude toward various types of parties and circumstances; and Democratic and Republican judges might differ in their approach to the judicial process, including in their views about the appropriate level of deference due to lower-court decisions. Democratic and Republican judges might even systematically differ in their personality traits and characteristics.

I find that the **evidence supports** the hypothesis that the **political affiliations of panel judges** can help to **predict outcomes in** a broad set of cases that together represent over **90% of circuit court decisions**. The association between political affiliation and outcomes is thus far more pervasive than has been recognised by prior research. To the best of my knowledge, my paper is the first to identify and characterise such a pervasive role of party effects throughout the large universe of circuit court cases.

B) ***Governments can't solve. Judge's lack of analysis leads to inconsistency. There is a circuit split, leading to confusion. That's American Legal Journal 24.***

2 Democracy D—2NC

No Democracy Impact — best studies show no correlation between democracy and peace, but instead democracies are more likely to attack other countries. That's Chiba & Gartzke.

Democracies are aggressive—this is true in all scenarios, including against other democracies.

Chiba & Gartzke 21 — Dania Chiba, Associate Professor of Political Science in the Department of Government and Public Administration at the University of Macau, Ph.D in Political Science from Rice University, LL.M in Jurisprudence and International Relations from Hitotsubashi University, Erik Gartzke, Professor of Political Science at the University of California, San Diego, PhD in Political Science from the University of Iowa, 2021 ("Make Two Democracies and Call Me in the Morning: Endogenous Regime Type and the Democratic Peace," *Office of Naval Research*, February 19th, Available Online at https://dainachiba.github.io/research/make2dem/Make2Dem.pdf, Accessed 09-07-2024)

We now turn to the results from the outcome stage, where militarized conflict initiation is regressed on democracy measures and other covariates. The univariate clog-log model 32 that ignores the endogeneity, shown in column (1) in Table 1, successfully replicates the standard, dyadic democratic peace finding that democracies are peaceful, though only toward other democracies. Note that, while individual democracy measures have either a positive or insignificant coefficient, joint democracy has a negative coefficient that overwhelms the positive coefficients of individual democracy measures in the univariate model. As a result, the univariate model produces a result that, while democracy may increase conflict against a non-democracy, it decreases conflict against a democracy.

To illustrate this, we calculate the average treatment effect of joint democracy for the challenger and for the target based on the univariate model. These effects are calculated by comparing the predicted probabilities of conflict initiation when changing the regime type of self (challenger or target) from non-democracy to democracy, holding constant the regime type of the other (target or challenger) as democracy. 33 Gray, hollow circles in Figure 4 show the treatment effects of challenger's and target's democracy. We can see that both effects are negative and statistically significant at the 95% confidence level.

Once we correct the endogeneity, however, the data no longer support such conclusions. In column (2) in Table 1, the negative coefficient for joint democracy no longer overwhelms the positive coefficient of challenger's democracy. Challenger's democracy now appears to increase conflict even against a democratic target. Red, solid circles in Figure 4 show the average treatment effects of challenger's and target's democracy, calculated from the trivariate model. The effect is positive and statistically significant for challenger's democracy, although the effect is indistinguishable from zero for target's democracy.

Whether we correct for endogeneity thus makes a significant difference in our estimates of the effect of joint democracy on conflict. The key to understanding why these changes occur lies in the estimated correlations between the error terms for different equations. The estimated error correlation between equations for conflict and challenger's democracy, 12, is negative and statistically significant. This suggests that unobservable or unmeasured determinants of a country's democracy make it less likely for that country to attack another country. A failure to control for such factors would generate a negative omitted variable bias, making it look as if challenger's democracy has a pacifying effect on conflict behavior. On the other hand, the estimated error correlation between conflict and target's democracy equations, 13, is indistinguishable from zero, suggesting that the endogeneity problem does not seem to operate for target's regime type.

Democratic peace theory is cherry-picked, and dyadic between two democracies at best.

Doorenspleet 19 — Renske Doorenspleet, Political Science and International Studies Professor at Warwick University, 2019 ("Democracy and Interstate War," *The Theories, Concepts and Practices of Democracy*, Available Online at https://link.springer.com/chapter/10.1007/978-3-319-91656-9_3#citeas, Accessed 09-07-2024)

This finding or 'law' has not only been recognized by scholars of international relations, but also found its way outside academia and has influenced foreign policies to promote peace and democracy, most prominently since the 1990s. However, my book will not draw conclusions based on '**cherry-pick**ing' of specific studies showing how peaceful democracies are, but on a systematic overview of studies in this field. Therefore, this book relies on **my own database** with **hundreds of different studies**, which are relevant for each chapter; the articles had to engage directly with the chapter's main research question. The next section will provide more detailed information around the selection criteria. This overview includes both highly cited and recent articles which were selected in a systematic way. Based on analyses of statistical studies around this topic of democracy and war, it will become clear that the overall statistical support for the democratic peace hypothesis is not strong at all. In the rest of the chapter, I will spell out four reasons why democracy does **not cause peace**, and why the **empirical support** for the popular idea of democratic peace is **quite weak**: (1) most studies do not find a **strong correlation** between democracy and interstate war at the **dyadic level**, and they show that there are **other—more powerful—explanations** for war and peace, or even that the impact of democracy is a **spurious one**, (2) the **theoretical foundation** of the democratic peace hypothesis is **weak**, and the causal mechanisms are **unclear**, (3) democracies are **not necessarily more peaceful** in general, and the evidence for the democratic peace hypothesis at the **monadic level** is inconclusive, and (4) the process of democratization is dangerous and living in a democratizing country means living in a less peaceful country.

In my view, it is difficult—if not **impossible**—to support the democratic peace hypothesis without any reservations. The key caveats should not be ignored and certainly deserve more attention before we can confidently argue that democracies are more peaceful than other types of political systems. Please notice that I can already reveal that the assumed link between democracy and intrastate war is problematic as well, but this topic will be at the core of the next chapter (Chapter 4).

Selection of Articles: Democracy and War

The instrumental value of democracy cannot convincingly be found in democracy's expected bond with peace. I have come to this conclusion on the basis of an analysis of statistical studies, which will be discussed in the rest of this chapter. So how did I select the articles for my database?4

For this chapter and Chapter 4, I selected the articles that focused on war and democracy. Using the online database Web of Science (formerly known as Web of Knowledge), I identified a total of almost **8000 articles** published in the sampled journals until the end of 2015. I identifed them by entering 'democr*' in the basic search field; this asterisk (*)-based 'wildcard' allows searching for terms including 'democratic', 'democracy', and 'democratization' (in both British and American spellings) simultaneously, in the title, abstract and/or the keywords. As a next step, I excluded articles in which 'democracy' is used as synonym for state (e.g. analysis of the relationship between immigration policies and unemployment in European democracies) or a specific political party or movement (e.g. the 'Democrats' in the USA, or Uganda's People's Democratic Army) or a specifc country (e.g. the Democratic Republic of Congo). In addition, I identified them by entering 'war*' in the basic search feld. The words democracy (democr*) and war (war*) need to be mentioned in title and/or abstract—and I also checked for equivalents of 'war' like 'conflict' and 'dispute' and 'no peace'.5

As it is not feasible to analyse thousands of articles, it is necessary to take a next step in the selection process. I decided to select these articles, which will be part of the database for the third and fourth chapter, in three different ways. The first method is to choose the articles with the most citations. So, for example, in Chapter 3, the articles which are cited more than a hundred times are included in this first list. The article by Beck et al. (1998) has been cited more than 951 times, and as a consequence, this article is part of the database. But also articles with a much lower number of citations (such as Barbieri 1996, with 179 citations) are included in my analyses.

The second method is simply to include the most recent articles published in the past five years, so since beginning 2011 until end 2015. The most recent articles can easily be overlooked by applying the first method of most quoted articles. In my view, however, they still need to be included as they present the most recent findings and engage with the recent and innovative debates, which cannot be ignored in this book. For example, recent studies on democracy and interstate war (Chapter 3) have paid more attention to the mechanisms (see, e.g., Zeigler et al. 2014), and there is a growing attention for the impact of political institutions in recent studies on democracy and intrastate war (Chapter 4; see, e.g., Walter 2015). Those recent findings cannot be ignored in any systematic analysis of statistical studies on this theme.

The third method is the most subjective approach of selecting articles, as it is based on the 'snowballing method'. So it includes articles which have not been selected by the first and second methods, but which have been quoted extensively and regularly by the previously selected articles. For example, the article by Bethany Lacina (2006) cannot be selected based on having high citations (the first method) and it cannot be included based on being a recent publication (the second method), but it has been mentioned by key studies and hence surfaces via the snowballing method (a third method). This article is important as it **clearly distinguish**es the determinants of **conflict severity** from those for **conflict onset**, and those determinants seem to be **quite different**, which is crucial information for Chapter 4.

In this way, my study presents and assesses the findings based on a big pool of statistical studies in the published literature. Based on this assessment, I will be able to draw clearer conclusions concerning the significance of the effects of democracy on interstate war (this chapter) and intrastate war (the next chapter). The Appendix shows more detailed information of the selected articles.

The Democratic Peace Hypothesis, Its Roots and Supporters

The democratic peace hypothesis6 states that democracies never or seldom go to war with one another. Where is this powerful idea of 'democratic peace' coming from? Before discussing the main findings of the statistical articles and before describing the four caveats of the 'democratic peace paradigm', we need to know a bit more around the background and the roots of this idea.

Immanuel Kant's 1795 essay Perpetual Peace has often been mentioned as the foundation for this hypothesis. Kant believed peace was difficult to achieve, since 'the natural state is one of war' (Kant 1795: 10). A state of peace must therefore be established for—in his view—it is certain that hostilities will be committed and people need to be protected from each other. In such a world, each may treat his neighbour, from whom he demands security, as an enemy. In a dictatorship where 'the subjects are not citizens, a declaration of war is the easiest thing in the world to decide upon, because war does not require of the ruler, who is the proprietor and not a member of the state, the least sacrifice of the pleasures of his table, the chase, his country houses, his court functions, and the like. He may, therefore, resolve on war as on a pleasure party for the most trivial reasons, and with perfect indifference leave the justification which decency requires to the diplomatic corps who are ever ready to provide it' (Kant 1795: 13).

In contrast, the situation is different in constitutional republics, according to Kant. He argued that the majority of the people in republics would never vote to go to war, except for pure self-defence. Therefore, a world with only republics would be peaceful, since there would be no aggressors. The republican constitution, which requires the consent of the citizens to start a war, gives the positive prospect of perpetual peace.

It is important to note that the ideas of Kant on the one hand and the modern democratic peace scholars on the other hand are not completely similar. For example, Kant talked about republics instead of democratic states as the ideal states to achieve peace. He defined republican states as states with representative governments, in which the legislature is separated from the executive. Not surprisingly—considering the epoch in which he lived—Kant did not include universal suffrage in his definition, which is now seen as an essential dimension of democracy, even of the most minimalist types of democracy (Dahl 1971; see also Chapter 2). Moreover, Kant argued that republics will be at peace in general, which means that such political systems are expected to be not only in peace with each other, but also with other non-republican systems. Nowadays, **only few scholars** would support this approach of a '**monadic democratic peace**'. As will become clear at the end of this chapter, there is **not much evidence** for the idea that democracies are **more peaceful in general**.

Since the 1960s, most statistical studies have not focused on the 'monadic democratic peace hypothesis' but on testing the 'dyadic democratic peace hypothesis'. This dyadic hypothesis states that it is less likely that democracies fight with each other, compared to other 'dyads' or other pairs of different types of political systems. The sociologist Dean Babst was the first scholar who started to build on Kant's old idea in the 'dyadic' way, and decided to test it in statistical studies (Babst 1964, 1972). He concluded that 'no wars have been fought between independent nations with elective governments between 1789 and 1941' (Babst 1972: 55). His study was not published in one of the journals in the field of international relations, but in a sociological journal and later in Industrial Research. Therefore, it was not read by international relations scholars, and initially, it did not get the attention it deserved in the field of international politics. Babst's work was, for example, not cited by Melvin Small and J. David Singer (1976), and their fndings seemed to contradict Babst's study. However, Small and Singer did not compare the rates of war proneness for democracies and dictatorships, but instead they focused on the question whether wars involving democratic states have historically been significantly different in length or in degree of violence compared to wars involving only dictatorships. For length and degree of violence during the wars, they did not find a difference between democracies and dictatorships, so they concluded that types of political systems did not matter.7 A few years later, Rudolph J. Rummel did cite Babst's work and replicated Babst's idea in statistical tests, which were described in the fourth book of his five-volume Understanding Confict and War (1975– 1981). He found clear support for his eleventh (of the 33) propositions about causes and conditions of conflict, which stated that 'Libertarian systems mutually preclude violence' (Rummel 1979: 279).

Eventually, those innovative studies from the 1970s helped to evoke the interest in the democratic peace proposition, and in the expected peaceful nature of relationships among democratic states. Since the 1980s, the number of quantitative studies has increased considerably, accumulating into an impressive field of research in international relations with its own 'empirical law' of democratic peace (Levy 1989: 270; see also Ray 1998).

This democratic peace hypothesis has not only received support from political scientists, but also from politicians and policy makers. Particularly since 1993, the idea of a democratic peace has inspired American foreign policies aimed at the promotion of peace and democracy. As the 42nd president of the USA (1993–2001), Bill Clinton was the first politician who explicitly bridged the gap between these findings in international relations on the one hand, and his foreign policy strategy on the other hand, at least rhetorically. Anthony Lake, who was Clinton's National Security Adviser, stated in 1993 that in order to cope with America's foreign policy challenges, the expansion of democratic states around the world would be essential because 'it protects our [U.S.] interests and security' (see Henderson 2002: 20). In his 1994 State of the Union, Clinton declared that 'Ultimately, the best strategy to ensure our security and to build a durable peace is to support the advance of democracy elsewhere. Democracies don't attack each other'.8 Findings from research in the field of international relations seemed to have a direct impact on policy making, and this move of the Clinton administration can be seen as 'a textbook case of arbitrage between the ivory tower and the real world' (Gowa 1999: 109).

Clinton's successor, George W. Bush, went one big step further in his faith that democratic peace holds. He argued that efforts to turn Iraq into a democratic country would have positive effects on Iraqi's neighbours. The authoritarian regimes in the region would fall as domino stones and follow the Iraqi example. They would start democratizing as soon as they could, which would then result in achieving a peaceful and stable the Middle East. The real motives for attacking Iraq may have been different, but 'regime change' was at the heart of Washington's rhetoric when the USA started to bomb Baghdad in March 2003. The rhetoric of the Bush administration focused on toppling Saddam Hussein's regime, and replacing the entire underlying dictatorial system with a democracy.

Moreover, George W. Bush used the democratic peace idea to justify the war in Iraq, declaring, 'The reason why I'm so strong on democracy is democracies don't go to war with each other…I've got great faith in democracies to promote peace. And that's why I'm such a strong believer that the way forward in the Middle East, the broader Middle East, is to promote democracy'.9 In 2004, the 43rd President of the USA said: 'If you think you can have peace without democracy – again - I think you'll find that - I can only speak for myself, that I will be extremely doubtful that it will ever happen'.10 In his second inaugural address, he stated that 'the survival of liberty in our land increasingly depends on the success of liberty in other lands. The best hope for peace in our world is the expansion of freedom in all the world'.

Again, these are just words from speeches and can hence be seen as rhetoric to defend military intervention (cf. Jervis 2003; Kaufmann 2004; Daalder and Lindsay 2005; Owen 2005; Lieberfeld 2005; Schmidt and Williams 2008). Still, in the end, politicians have rationalized their political decisions based on one of the most powerful ideas taken from studies in the field of international relations, clearly showing the influence of this academic idea in political practice.

Hence, the field of international relations seems to have its own law: democracies rarely fight with each other. It cannot be denied that the evidence supporting the democratic peace proposition is quite diverse in character (see Ray 1998): the evidence has been epistemological (Rummel 1975), philosophical (Doyle 1986), formal (Bueno de Mesquita and Lalman 1992), historical (Weart 1994; Ray 1995; Owen 1994), experimental (Mintz and Geva 1993), anthropological (Ember et al. 1992; Crawford 1994), psychological (Kegley and Hermann 1995), economic (Brawley 1993; Weede 1996) and political (Gaubatz 1991). Still, there have been numerous critical studies (see, i.e., Hayes 2011), and the general picture is unclear. We do not yet know much about the overall findings from statistical studies.

So far, it seems as if some quantitative studies—mainly within the field of international relations—have found strong and robust evidence which supports the 'democratic peace hypothesis'. Those studies show that democracy has had a positive influence on international peace (see, i.e., Rummel 1979; Ray and Russett 1996). In this sense, the idea of a democratic peace seems to be confirmed. Political scientists such as James Lee Ray are passionate supporters: 'No scientific evidence is entirely definitive' but 'based on all the empirical evidence so far' the more defensible of the two possible definitive answers to the question "Does democracy cause peace?" is "Yes" (Ray 1998: 43). However, based on my own analyses of the empirical studies with statistical evidence, I cannot be as enthusiastic as those scholars; to the contrary, I whole-heartily disagree with them, as a more systematic analysis of the articles shows that there are four important weaknesses, which seriously undermines the idea that peace is one of democracy's instrumental values.

Caveat 1: It's Not (Just) Democracy

While analysing the selected articles, the first remarkable finding is that only a relatively small number of studies have actually tested the democratic peace hypothesis. Most of the studies have focused on the mechanisms (see next section, caveat 2), and hence seem to assume that there is a correlation between democracy and war. In this way, the majority of the studies—often unintentionally—reinforce the idea that democratic peace actually exists without testing this proposition. However, **no**ne of the **studies** that **directly test** the democratic peace hypothesis found **strong evidence** that democracy is the **most important factor** when explaining interstate war. All democratic peace studies have controlled for many possible alternative causes of the peace, such as economic development and growth, geographic distance and contiguity, power status, alliance ties, militarization and political stability. The findings show that it is not just democracy which explains war, not at all. Within this group of studies, which explicitly test the democratic peace hypothesis, four different types of findings can be detected. I will discuss those results more in-depth in the rest of this section.

First Result: There Is Correlation, but Other Explanations Are Significant Too

The first subgroup consists of scholars who stress the importance of democratic peace, despite the fact their own analyses have shown that other factors are statistically significant as well (Maoz and Russett 1993; Rousseau et al. 1996; Gleditsch and Hegre 1997; Beck et al. 1998; Ray 2013). For example, some studies (e.g. Rousseau et al. 1996) included alternative independent variables in order to test realist arguments. They tested whether the distribution of power determines decisions to use force, and measures each state's military capabilities relative to its opponent. A state's military capability is the average of three elements: number of troops, military expenditures and military expenditures per soldier. They found that this realist variable was strong, positive and statistically significant at the 0.001 level in their analyses (see, e.g., Rousseau et al. 1996: 522, Table 2). However, not only a state's military capabilities appeared to be an important explanation for peace. In addition, wealth, growth, alliances and contiguity played a crucial role when explaining interstate war (see, e.g., Maoz and Russett 1993: 632, Table 1).11 Moreover, when **other factors** are included, the **impact of democracy** on the likelihood of international crises is **even spurious** (Maoz and Russett 1993: 632; Henderson 2002: 141, see also p. 3).12 Still, scholars in this group keep defending the democratic peace idea, despite the fact that their own analyses showed the significance of alternative explanations.

Second Result: Initially There Is Correlation, but the Impact of Democracy Is Spurious When Other Explanatory Factors Are Included in the Models

The second subgroup of scholars is far more radical. Based on their own analyses, this group concludes that the democratic peace link is a spurious one (Weede 1984, 1996; Barbieri 1996; Mousseau 2013; Gartzke and Weisiger 2014).13 Typically, efforts to demonstrate the spuriousness of the statistical democratic peace pointed to other factors that, when accounted for '**properly**', **eliminated** or dramatically reduced the statistical significance of shared democracy. Hence, the studies in this second group did not find strong evidence for the democratic peace hypothesis anymore, once other explanatory factors were included in the models.14

One of the most convincing alternative explanations of peace between countries is that there is **no democratic peace**, but a **capitalist peace instead**. The settlement in Germany and Japan succeeded because of the establishment of capitalist peace. Because of economic support by the Americans, who encouraged free trade and offered trade opportunities in practice as well, the poorer economies in Europe and Japan would gain economically, resulting in 'economic growth, prosperity, and, ultimately, free trade among most of the more technologically advanced economies' (Rasler and Thompson 2005: 232). By establishing and expanding free trade, the incentives for war would quickly decrease among trading states, according to this approach. To prevent new interstate wars after World War II, the capitalist peace was a far more important factor than the American promotion of democracy and its political institutions.

The capitalist peace, or capitalist peace theory, also states that economic development accounts for both democracy and the peace among democratic nations. Economic development is a key factor to explain democracy (Lipset 1959; see also Hegre 2003; Weede 2004).15 Moreover, economic development also plays a role when explaining peace, and the presence of market-oriented economies in countries have a positive impact on both democracy in those countries and peace between them (Mousseau 2000, 2002, 2003, 2005, 2013; see also Hegre 2014).

Democratic peace **only exists** when both democracies have **high levels of economic development**, when economic development is well above the global median.

In fact, the poorest 21% of the democracies studied, and **the poorest** 4–5% of current **democracies**, are **significantly more likely** than other kinds of political systems to **fight each other** (see, e.g., Mousseau 2005). Moreover, if at least one of the democracies involved has a very low level of economic development, then democracy cannot prevent war.16 Still, there is a pacifying effect of free trade and economic interdependence, which is **more important** than the effect of democracy, because the former affects peace both directly and indirectly, by producing economic development and ultimately, democracy (see Weede 2004).17

Capitalist peace is not the only alternative explanation. Shared interests in general, and political similarities in specific, can also be seen as an important second alternative explanation for war and peace between countries (Farber and Gowa 1995, 1997; Gartzke 2007; Gowa 1999; Henderson 2002). Democracies are **not peaceful** to each other because they are **democratic**, but rather because they are **similar**. So the difference of the scores of both countries also contributes to the conflict proneness of the dyad. If the difference in levels of democracy is big, then the chance of conflict is higher (cf. Oneal and Russett 1997: 281–282).

3&4 Patents Deck Innovation—2NC/1NR

Extend patents deck innovation on previous ev: A) Downstream Blocking — patents raise input prices and make impossible for people to improve innovations. B) Patent Trolls — litigation costs incentivize patent trolls. Companies only use patents as a defense to legal disputes. C) First-Mover Advantage Solves — the incentive of market control is sufficient to solve. That's Boldrin and Levine.

1. Monopolies — they create high prices and deck follow-on innovation.

Osborn et. al 15 — Lucas Osborn, Professor of Law at the Norman A. Wiggins School of Law, Campbell University, North Carolina, Joshua M. Pearce, B.Sc. in Physics and Chemistry, The Pennsylvania State University, Ph.D. in Materials Engineering, The Pennsylvania State University, John M. Thompson Chair in Information Technology and Innovation at the Thompson Centre for Engineering Leadership & Innovation, Amberlee Haselhuhn, Senior Director of Technology at LIFT, 2015 ("A Case for Weakening Patent Rights," *St. John's Law Review*, Vol. 89, Winter 2015, No. 4, Available Online at https://s3.amazonaws.com/real.stlouisfed.org/wp/2012/2012-035.pdf, Accessed 08-23-2024)

Regardless of the theory to which one ascribes-the incentive to invent view, the prospect view, or variants thereof-the patent system unfortunately imposes key costs on society. First, by giving an exclusive right to its owner to make, use, sell, and offer to sell the invention, a patent raises the potential for the inventor to sell the invention at a price higher than what it would command in a perfectly competitive market. To the extent there are no reasonable substitutes, a patent holder can charge a higher monopoly price for the invention and thus make more profit per item sold. The increased price forces some purchasers out of the market for the item, creating a deadweight loss.

Second, the patent system can also burden society by impeding follow-on technology.' Technology creation is cumulative; inventors build on the inventions of yesterday to bring forth new inventions. Patents can discourage follow-on research by preventing the inventor of an improvement from commercializing it to the extent that it infringes the first patent.10 The longer technology remains patented, the slower the cumulative research advances that build upon it will be.

2. Litigation Costs — they suck up resources for innovation, costs billions.

Baker 16 — Dean Baker, co-founded of the Center for Economic and Policy Research, holds a Ph.D. in Economics from the University of Michigan, 2016 ("Rents and Inefficiency in the Patent and Copyright System: Is There a Better Route?" CEPR, August, Available Online at https://www.cepr.net/report/working-paper-rents-and-inefficiency-in-the-patent-and-copyright-system-is-there-a-better-route/, Accessed 08-21-2024)

They cite a wide range of evidence that patents can be a major source of waste and a **hindrance to productivity growth**. For example, the vast majority of patents are never used. Old established companies often stockpile patents to use as competitive weapons against smaller upstarts. Hall and Ziedonas (2001) examined the upsurge of patents in the semi-conductor industry in the 1980 and 1990s. They found that the main motivation was to use patents as a weapon in lawsuits against competitors, as well as bargaining chip to be used in a settlement in cases in which they were the defendant. Because litigation involves large costs, an established firm is much better situated to contest a patent than an upstart with few resources. As a result, a patent can be used to force the upstart to share much of the benefits of its technology, even if there is no actual dependence on the patent of the established firm. This sort of reasoning was commonly cited as the main explanation for Google's decision to buy the Mobility division of Motorola in 2011 for $12.5 billion. At the time, as a relatively new company, Google did not have a large portfolio of patents which could be used as a retaliatory weapon when it was sued. The purchase of Motorola's Mobility division gave Google a large portfolio of patents that could be used in retaliatory suits against competitors.

The extreme case of using patents for legal harassment is that of an actual patent troll, a company that **only exists to push claims to patent rights against profitable companies**. Boldrin and Levine (2013) note the case of NTP, a patent troll that won a patent infringement case against Research in Motion (RIM) over the Blackberry. In order to avoid having its system shut down at a point where its service was expanding rapidly, RIM agreed to pay NTP $612.5 million to license the use of the patent. On appeal, the original ruling was overturned; however, RIM did not get its money back. The implication is that more than $600 million was taken from, what as at the time a thriving and innovative company, due to a mistaken judicial ruling. Of course, this ruling provided an enormous incentive for other companies to try to follow NTP's example.

One recent study by Bessen and Meurer, which relied on a survey of corporate executives, put the direct cost to firms of litigation with patent trolls (including settlements) at **$29 billion** for the year 2011. 9 An earlier study involving the same authors, which looked at the impact on stock prices, put the cost at **$80 billion a year.**10 Most of the cost in these estimates stems from payments made to the patent trolls or the need to alter their business plan in response to a patent suit. Insofar as these payments reflect compensation for legitimate innovations (a claim disputed by Bessen and Meurer), they would not constitute rents associated with the patent system, they would simply be redistributions among patent holders. But even with this generous interpretation, Bessen and Meurer still attribute more than $5 billion of their $29 billion estimate to direct litigation costs.

These litigation costs are pure waste from an economic standpoint. The actual waste to the economy would have to be several times this size, because the patent trolls undoubtedly spent a comparable amount on litigation. In addition, this study is only looking at suits with patent trolls (formally, nonpracticing entities (NPEs)). These suits account for roughly 60 percent of all patent suits. While suits brought by companies that actually use the technology may be more meritorious on average, the legal expenses are still a cost to the economy. Extrapolating from the $5 billion costs estimate of litigation costs, total litigation costs related to patent for 2011 could have easily been close to $17 billion or **7.3 percent of total R&D spending for the year.**11 And this does not even account for the extent to which payments resulting from these suits may not be merited, as was the case with the NTP suit and which Bessen and Meurer argue is the case with most suits involving NPEs.

Boldrin and Levine (2013) also note the substantial legal costs associated with patent protection. It points out almost 250,000 patents were filed in 2010. It puts average legal costs at more than $7,000 per patent, implying that firms spent almost $2 billion in legal fees in 2010 just to file patents. Furthermore, with the ratio of litigation to patents remaining roughly constant, while the ratio of patents to R&D spending has risen considerably over the last three decades, the ratio of litigation to R&D spending has clearly increased. From the standpoint of the economy, these additional legal costs are a pure deadweight loss.

The legal issues surrounding the **proliferation of patents can obstruct innovation** in a variety of ways. Shapiro (2001) notes the problem of "patent thickets," situations where innovations often involve the use of a large set of patents. This can result in large transactions costs, which may stifle innovation. The problem can be even more serious if inadvertent infringement can result in major penalties. The paper notes that the problem of patent thickets has become especially serious in several important sectors, such as semiconductors, biotechnology, computer software, and the Internet, since all have seen a proliferation of patents in recent years. In the same vein, patents on research tools, such as transgenic animals and biological receptors, have become increasing common in the last three decades. The royalty payments and transactions costs associated with these tools can make the research to develop new drugs and medical diagnostic products considerably more expensive and thereby slow the process.

3. Startups — threat of litigation forces startups out of industries.

Baker 16 — Dean Baker, co-founded of the Center for Economic and Policy Research, holds a Ph.D. in Economics from the University of Michigan, 2016 ("Rents and Inefficiency in the Patent and Copyright System: Is There a Better Route?," CEPR, August, Available Online at https://www.cepr.net/report/working-paper-rents-and-inefficiency-in-the-patent-and-copyright-system-is-there-a-better-route/, Accessed 08-21-2024)

Recent research has also found considerable evidence that the threat of patent litigation **distorts** the **direction** of research and is a powerful **weapon** of larger firms against smaller firms and start-ups. Lerner (1995) examined the patenting behavior of biotech firms. It found that firms facing higher **legal costs**, due to their small size, are **less likely** to patent in subclasses where there are many other patents. This is especially likely if the firms holding the other patents in the subclass are larger firms with substantial legal resources.

Similarly, Lanjouw and Schankerman (2001a) found evidence of a strong reputation effect where **patent holders** are more likely to file suits in areas where many new patents are being issued. This presumably means that companies want to **show** their **willingness** to contest patents to **intimidate** competitors. Suits were also more likely in patents with fewer backward citations. It takes this as evidence that in new areas where the bounds of existing patents are less well-established there will be a larger basis for contesting claims.

Both of these **findings** are **troubling** from the standpoint of **promoting innovation**. Insofar as a reputation effect is important for being able to protect a claim, it means that larger firms will be better **situated** than smaller ones who may have difficulty covering the cost of litigation. The finding that patent suits are more likely in new areas implies that litigation will more frequently be needed to protect patents that are opening new ground. Here also the implication is that patents will be of less value to **smaller upstarts** than **well-established firms.**

Lanjouw and Schankerman (2001b) found that **smaller firms** and **individual holders** are far **more likely** to be involved in patent suits than large firms. The disproportionate negative effect on startups is made worse by the fact that large patent portfolios seem to provide **protection** from suits. Firms with large patent portfolios are less likely to be involved in patent suits even when controlling for size. The study finds that the probability of a suit going to trial or the settlement size is not affected by firm size. The conclusion of this analysis is that litigation costs are **greater to smaller firms** because they are less well-situated to pursue litigation **avoidance** strategies. This means that patents are a less valuable asset to smaller firms because they are **more costly** on average for them to enforce.

Lanjouw and Lerner (2001) found that larger firms were 16 to 25 percent more likely to gain a preliminary injunction in a patent suit than **smaller firms**. This figure likely understates the bias in favor of large firms because lower litigation costs would mean that they would be more likely to pursue weak patent claims than smaller firms. This indicates a substantial tilting towards large firms because a preliminary injunction is an important weapon in a suit, granting the plaintiff the opportunity to effectively maintain a monopoly in the market for the duration of the injunction, while preventing the defendant from receiving a return on its investment.

4. Empirics — best evidence proves patents hinder innovation.

Hicks 24 — James Hicks, J.D. and Ph.D. in Jurisprudence and Social Policy from the University of California, Berkeley, B.A. in Political Science from Reed College, Portland, Oregon, Academic Fellow at Columbia Law School, Fellow in Law, Business, and Innovation at Berkeley Law, 2024 ("Do Patents Drive Investment in Software?," *Northwestern University Law Review*, March 10th, Available Online at https://scholarlycommons.law.northwestern.edu/cgi/viewcontent.cgi?article=1559&context=nulr, Accessed 08-22-2024)

IV. RESULTS: DO PATENTS AFFECT INVESTMENT OUTCOMES?

In this Part, I turn to my central question: do patents drive investments in startups? I begin by examining whether, and to what degree, the Alice decision impacted patent approval rates. Next, I use these findings to analyze the impact of patents on the startups' success on two fronts: first, the incidence and amount of investment, and second, the likelihood that a startup was acquired or launched an initial public offering. Despite the suggestive evidence in the prior literature, I find no evidence that patent grants drive either venture capital investments or later success.

A. First Stage: Did Alice Matter?

The first step is to estimate the impact of Alice on the treatment. In other words, if Alice had an effect on patent examination for business-methods inventions, we would expect to see a decline in allowance rates following the decision. Figure 1 shows the average allowance rate organized by date of final disposition over a window of two years before and after Alice. 126 Each point reflects the average rate over approximately two weeks of dispositions. The difference is immediately apparent: average allowance rates for firsttime business-methods applicants fell from 45% to around 10% in the months following Alice. The decision had an unambiguous—and very fast— impact on business-methods applicants' chances of receiving a patent.

[Graph Omitted]

B. Funding

Having established that Alice caused a sharp decline in patent grants, I turn to outcomes. If it is true that patent ownership affects investment decisions, then we would expect to see a corresponding drop in the share of companies that received venture capital funding after Alice. To test this, Figure 2 shows the rate of post-disposition funding for applicants, again grouped into two-week bins. Although the average likelihood of funding is somewhat noisy, ranging from 0%–30% of companies in each month, there is no evidence of any change in investment, discontinuous or otherwise, in the wake of the PTO's new Alice standards. Indeed, the rate of funding appears to be steady throughout the four-year window, regardless of when an application was finally decided.

[Graph Omitted]

Perhaps, however, the difference is not in the rate of funding, but in the amount invested? Although I don't observe any change in the likelihood of investment, it's possible that startups whose patent applications are granted raise more money in absolute terms. To test this alternative, I narrow my focus to the subset of startups who received at least some funding after the disposition of their first application. In Figure 3, I plot the average total investment for these companies, once again grouped by the two-week period in which their application was decided. Figure 3 shows that, amongst companies who received post-decision investment, the average amount invested ranged between $1 million and $100 million. But the graph also tells the same story as with the incidence of funding: there's no evident change in the average amount of funding after Alice. 127 I find no evidence that a patent grant leads to a change in the average amount invested in startups.

[Graph Omitted]

In the remainder of this Section, I repeat these analyses with more formal statistical tests. But the graphical results convey the core story: despite a precipitous fall in allowance rates for business-methods patents, the likelihood of funding and the amount invested remained unchanged compared to the companies whose applications were resolved before Alice. Again, I find no evidence that patents affect investment in business-methods startups.

Table 2 provides formal tests of Figures 1 and 2. The four columns show results across a range of different specifications and estimation windows. In practice, the implementation is as follows: I choose various numbers of months before and after the Alice threshold, and then fit a regression line on each side. The difference between these curves at the point of the Alice decision forms the estimate. Column (1) shows the results for a linear regression fit across the full pre- and post-Alice windows. Columns (2)–(4) show the results of difference-in-means estimates for windows of six, twelve, and twenty-four months. The first row, which is analogous to Figure 1, shows the results of a regression of patent approval on post-Alice disposition. If Alice diminished patent allowances, we should see a strong and negative relationship—and indeed we do: on the order of negative 30 percentage points. The second row, which is analogous to Figure 2, shows results of a regression of investment likelihood on post-Alice disposition.128 Again, if patent ownership drives investment, we should expect to see a drop in the likelihood of funding amongst the post-Alice group, who are much less likely to hold a patent. But here, on the other hand, the effect is approximately zero. Note, though, that this row is an estimate of Alice's effect on the chance of receiving investment—that is, it compares companies who received their disposition before the decision with those who received it after. Because not every company received a patent before Alice (and vice versa), we need to make a further adjustment in order to estimate the effect of a patent grant

[Table Omitted]

In Table 3, I show the results of the fuzzy RD, which uses the estimates above in order to zero in on the role played by patents.129 Again, the columns contain the results of several different regression models. In every case, the dependent variable is the same: whether the startup company received any venture capital equity investment following the first decision on the company's patent applications. The first row—"patent approved"—reports the key result: it shows the effect of obtaining a patent on receiving postdisposition investment.

The first column, labeled "OLS," provides some context: it reports the simple relationship between the presence of post-disposition funding and approval of a company's first patent application.130 This is a replication of the correlational findings in the prior literature, which are described in Section II.B. As expected, I find a strongly positive and statistically significant relationship. The coefficient implies that having the first patent application approved is associated with a 6.1 percentage-point (roughly 40%) increase in the likelihood of subsequently receiving funding. But we know from the earlier discussion that this estimate is biased in the absence of other information about the quality of the invention, the team, and so on.131

Columns (2)–(5) show the results of the (fuzzy) regressiondiscontinuity approach. Each column uses a slightly different specification, but the goal is always the same: to estimate the effect of the first patent approval on the likelihood of subsequent investment. As above, columns (2)–(5) show results for a range of different specifications and windows.

The results in columns (3) and (4) indicate an effect of approximately zero, while columns (2) and (5) are actually slightly negative. 132 But more importantly, none of these RD estimates are significantly different from zero.133 This accords with expectations from Figure 2 and Table 2 above: as those figures suggest, there is no evidence of a change in funding outcomes at the point of the Alice decision. This flows through to the fuzzy RD estimate: the results show no evidence that patent approval has an effect on obtaining funding for business-methods startups.

[Table Omitted]

One drawback of the conventional RD approach is that its estimates can be noisy, particularly with small sample sizes, which leads to the need to extrapolate by using data far from the threshold. A natural robustness check uses the local randomization interpretation of RD.134 Local randomization requires a stronger assumption about the research design, but in return offers hypothesis testing that is reliable in small samples.135 In Tables 4 and 5, I show the results of tests of the sharp null hypothesis of no difference between the rate and amount of funding before and after Alice. 136 In line with the results above, I cannot reject the sharp null hypothesis of no effect, regardless of the window of observation.

[Table Omitted]

C. Acquisitions and IPOs

Of course, the amount of investment is not the only way in which a patent might affect the success of a startup company. If a patent grant increases the latent value of a company—whether by signaling team competence, the stage of product development, or the existence of a legal monopoly—we might also expect the company to be a more appealing target for acquisition or an initial public offering.137 Because most startups fail, these events are in a sense the true outcome of interest. Capital raised is a proxy for success, but what investors seek is a profitable "exit" from their investments. The simple comparison of successful and unsuccessful patent applicants in Table 1 suggests that there is a modest difference in postdisposition acquisition activity: 14.6% of startups whose first patent was granted were subsequently acquired versus only 11.6% for unsuccessful companies. But once again, there's good reason to think that this simple relationship doesn't tell the full story in the absence of more information about the quality of the inventions, the product teams, and so on. We can use the same approach as in Section IV.A to test whether patent approval has a causal impact on exit outcomes.

Figure 4 shows the percentage of startup applicants who were acquired or entered into a merger after the disposition of their first application.138 Recall that Figure 1 revealed a sharp drop in patent grants after Alice was decided. Just as before, if it's true that patents drive acquisition activity, then we would expect to see an accompanying decline in that activity immediately following the decision. But once again, there is no evident effect at the threshold. Although acquisition activity ranges between 0% and 20%, the overall trend is steady regardless of when the application was decided. Tables 6a and 6b show the results of formal statistical tests, which support the null result.

[Table Omitted]

At the same time, the mere fact of acquisition may hide important variation. Not all startup acquisitions are alike: some are essentially fire sales—liquidations by another name—while others are true value-enhancing purchases.139 If it's the case that acquisitions continued at the same rate after Alice, but that they changed in kind, then Figure 4 might miss an important effect. To check for this, I subset the companies to those that experienced a "successful" exit event after the decision on their application. I define success as either launching an IPO or being acquired for more than the sum of previously raised capital.140 Because relatively few startups actually experience such an exit, the number of observations is much lower, and the graphical presentation would be very noisy. Instead, Table 7 shows the results of the same permutation test as before and also includes an estimate of the average difference between the rates of successful exit before and after Alice. Once again, as the large p-values indicate, I cannot reject the null hypothesis of no difference between "successful" downstream outcomes for the two groups of applicants.

In summary, there is no evidence that Alice has an effect on exit outcomes for business-methods software firms. Neither the likelihood of being acquired nor the chance of a successful exit appears to be driven by the approval of a firm's first patent.

5. R&D A) Holders — models proves patents discourage investment in research… the patent holder has no incentive to do research because they already have the monopoly and B) Competitors — Other companies have no incentive to do research in improving that product because they would not be allowed to sell it. That's 1NC Stallman

5 US Wins Tech Race—2NC/1NR

The US Dominates the Tech Race:

A) Talent — We hold half of high-skill immigrants, that ensures innovation.

B) Financing — US funding only grows. China's held back by sanctions, shortages, and inexperience.

C) <u>Market Economy</u> — US horizontal innovation makes it impossible for China to win even if they outpace in one area.
That's Sheng and Geng and Lewis.

<u>Patents are a bad metric</u> — far too many other factors and doesn't quantify Chinese innovation.

Lewis 24 — James Andrew Lewis, Senior Vice President, Pritzker Chai, Director of the Strategic Technologies Program, 2024 ("America is Losing the Shoe Race With China," *CSIS*, April 1st, Available Online at https://www.csis.org/blogs/working-papers/america-losing-shoe-race-china, Accessed 09-04-2024)

There is no easy way to directly measure innovation, so there can be a tendency to use proxies like number patent issued, amount spent on R&D, number of PhDs and publications. It seems reasonable to assume that more R&D spending, more publications, and more patents correlate with more innovation and one of their attractions is that they are easier to count, but by themselves they do not explain strength in technological innovation. Until the last few years, <u>the United States lagged in investment in basic research</u> (the foundation of innovation) and in the infrastructure to support innovation. <u>Yet it has over eighty years **outperformed other economies** when the measures are the creation of valuable new tech</u>nologies and national income. <u>The strength of the U.S. "innovation machine" lies somewhere else</u>. The discussion of technology and innovation relies on indirect proxy measurements, like the number of patents. <u>Patents are a metric that **must be used carefully**</u>. The assertion that "when a measure becomes a target, it ceases to be a good measure" is particularly true <u>for China</u>'s centrally directed economy, where people perform to meet measures set by Beijing, leading to situation where <u>researchers are **rewarded for the number of patents** issued</u> (or number of publications) <u>even if no one uses those patents.</u>

Many quantitative measures may not actually measure technological leadership as it relates to national power. Specific metrics such as the percentage of national income spent on R&D or <u>the number of patents</u> issued, <u>are inadequate by themselves to predict tech</u>nological <u>leadership</u>. <u>It is the **ability to use technology for commercial and military purposes** that is essential</u>. <u>Looking at the U.S. economy, it is investment in R&D, access to skilled labor, supportive business and intellectual property laws, a dynamic financial system, and an entrepreneurial business culture that explains its overall strength</u> compared to other economies.

<u>China Is Falling Behind in Innovation</u> — geopolitics, idea creation, entrepreneurs, investors, researches all causing decline.

Lewis 24 — James Andrew Lewis, Senior Vice President, Pritzker Chai, Director of the Strategic Technologies Program, 2024 ("America is Losing the Shoe Race With China," *CSIS*, April 1st, Available Online at https://www.csis.org/blogs/working-papers/america-losing-shoe-race-china, Accessed 09-04-2024)

<u>China is good at manufacturing</u> what others have invented <u>but **no longer so good at innovation** itself</u>. This is the result of political change. China was becoming a leading innovator when it was politically

open, before 2012. It still has advantages, but now that it is becoming politically closed, idea creation has slowed, **entrepreneurs, investors and researchers are leaving**, foreign investment is in decline (because of a perception of increased political risk), and geopolitics frays connections to global research and tech. Under different political leadership, China would be a much more formidable competitor, but China made a political decision that values continued party control over innovation and its ability to innovate is at risk.

Similarly, the EU spends significantly on R&D and has excellent research facilities, but its regulations are a powerful disincentive to entrepreneurship and commercialization. Europe's major economies have shown flat income growth for more than a decade. It is not positioning itself to compete in a digital economy, since this requires a willingness to accept risk and allow entrepreneurship. Europe values privacy over innovation.

The US leads in GAI investment. That spills over to all innovation.

Thornhill 24 — John Thornhill, Innovation Editor at the Financial Times, founder and editorial director of Sifted, founder of FT Forums, 2024 ("The great American innovation engine is firing again," Financial Times, May 9th, Available Online at https://www.ft.com/content/0d39e8f0-38ba-40aa-8ec8-d04e82afb690, Accessed 08-18-2024)

There is, of course, only so much that Washington can do. But the US **private** tech sector is independently enjoying a **surge** of new **funding** as investors bet big on the transformative power of **a**rtificial **in**telligence. US companies, led by Google, OpenAI, Nvidia, Microsoft and Anthropic, already dominate the field of generative AI. Goldman Sachs estimates that AI-related investment could rise to between **2.5 per cent** and 4 per cent of GDP in the US, compared with 1.5 per cent to 2.5 per cent in other leading economies.

"This is a **genuine breakthrough**," says Erik Brynjolffson, director of the digital economy lab at Stanford University.

Not only will **g**enerative **AI sharpen** the **competitiveness** of US companies, it will boost the **overall economy**, too. While the Congressional Budget Office is forecasting average annual productivity growth of 1.4 per cent over the next decade, Brynjolffson predicts it will be closer to 3 per cent, mostly thanks to AI. "It will roughly **double** the rate of **productivity growth**," he tells me.

Prefer Revenue and Market Share Metrics — they are the best predictors of innovative success.

Lewis 24 — James Andrew Lewis, Senior Vice President, Pritzker Chai, Director of the Strategic Technologies Program, 2024 ("America is Losing the Shoe Race With China," *CSIS*, April 1st, Available Online at https://www.csis.org/blogs/working-papers/america-losing-shoe-race-china, Accessed 09-04-2024)

These issues are not binary – an outcome where China makes all the shoes, and the United States makes none. We would need to consider both the number of shoes produced and their quality and type, and then ask how shoes contribute to wealth, power, and military strength and whether there are alternative technologies that can substitute for them (since under duress, nations are inventive). Answering these questions requires a broader view than that taken in many reports that looks at the overall economy and its technological base. The best predictors remain overall **national revenue and tech market share**.

The keys to technological leadership are at the most fundamental level, access to capital and to ideas (both the ability to create new ideas and to use them). Better measures of success include market share (meaning someone actually buys what you are making), the number of startups (particularly 'unicorns'), revenue, and the long-term trend in national income. These revolve around the central importance of the ability to commercialize research and innovation. This is an area – entrepreneurship - where the United States has had an advantage over Europe and China, both of which score well on proxy innovation metrics but less well for outcomes. The biggest risks to the U.S. innovation system are political dysfunction and regulatory ideologues. These are domestic problems, not the result of international competition, but is easier to fault China than in the mirror.

US advantage in deep tech companies ensures the tech lead.

Sanandaji 24 — Nima Sanandaji, Director at the European Centre for Entrepreneurship and Policy Reform, 2024 ("North America Dominates Deep Tech, But WIll It Last?," New Geography, May 16th, Available Online at https://www.newgeography.com/content/008172-north-america-dominates-deep-tech-but-will-it-last, Accessed 08-18-2024)

A key finding is that fully 72 percent of the leading deep tech companies are based in North America, **68 percent** in the **US alone**. This compares with 14 percent in Europe, **11 percent** in Asia. Africa, Oceania, and Latin America each host around one per cent of the leading deep tech firms. Santa Clara Valley, or Silicon Valley, is the leading deep tech hub globally. About 24 per cent of all the leading technology firms reside here. Thomas Edison founded the world's first industrial innovation laboratory in this valley 150 years ago, and it has since become the most significant region for development of novel technologies. The second most important global hub for deep tech is **New York**, where close to 8 percent of all globally leading deep tech companies are located. Boston and Los Angeles are other important hubs. Outside the US, the leading hubs are London, Tel Aviv, New Delhi, Toronto, Paris, Tokyo, Bengaluru, Amsterdam, Berlin, Stockholm, Montreal, and Dublin.

North America has particularly strong dominance in the areas of quantum & computing, as well as in pharmaceuticals and artificial intelligence. In these three deep-tech fields, four out of five or more of the leading global firms are in North America, predominantly the USA. Quantum & computing includes those companies that work on new quantum computers, as well as traditional computer technology. In this field, 44 out of the 50 leading global companies are found in the USA, and additionally one in Canada, which means that fully 90 percent are in North America. The Santa Clara Valley alone has a majority of 26 out of the 50 leading companies in quantum & computing.

In the field of pharmaceuticals, 41 out of the 50 leading global companies are found in the USA, and additionally a couple in Canada, which means that 86 percent are localized in North America. The Boston urban region has fully 13 out of these companies, with a further seven in New York, three in Santa Clara Valley, and three more in San Diego.

The third field with a particularly strong presence for North America is artificial intelligence. In this deep-tech field, 37 out of the globally leading 50 companies are localized in the USA and a further 3 in Canada. Thus, 80 percent of the leading companies are found in North America. Most of the leading artificial intelligence companies are found in the Santa Clara Valley, 27 out of 50, with New York hosting additionally three.

[charts omitted]

In fact, **in all deep-tech areas** except Clean Tech, **more than half** of the leading **top-50** firms of each field is located in the **USA**. Canada is however strong in Clean Tech, why even in this area half of the globally leading deep-tech companies are located in North America. The share of North American deep-tech companies is lowest for Clean Tech (50%) and highest for quantum and computing (90%).

**Top companies and universities prove the US is ahead.*

Anjum 24 — Meerub Anjum, Data Journalist at Insider Monkey, 2024 ("25 Most Innovative Economies in 2024: Why WIPO Rankings Are Inaccurate," Yahoo Finance, January 31st, Available Online at https://finance.yahoo.com/news/25-most-innovative-economies-2024-203515105.html?guccounter=1&guce_referrer=aHR0cHM6Ly93d3cuZ29vZ2xlLmNvbS8&guce_referrer_sig=AQAAAG-K004toGGnW3gFwXlz6nfGGzJUp6bUM79G1LXDLcmVlWXbLdA4okSfyYm0hsTW5Quzmt6l6FAs7DbZuGuNxwlOyKn7snsv9q9L2bq-AgjrZoIxiSAoHW6qEC2k8ksZJ6GNrEqUkKtCGikIkuGGhgKvJflc7WcaW8htvF2J9_IG, Accessed 08-18-2024)

The GII ranks Switzerland as the most innovative country in the world. While Switzerland has made significant progress in science and technology in the past years, the United States is **undoubtedly** the **most innovative economy** in the world. As of 2023, the US boasts a GDP of $26.95 trillion and is a leader in science and technology. **More than 75% of the most innovative companies** in the world are **based** in the **US**. The country is home to some of the **top universities** and research institutes including **MIT**, **Harvard**, and **Stanford University**, among others. The United States also stands as a global leader in sectors including health, finance, and energy, among others. The US is tentatively the biggest research and development **spender** in the world. It spent a total of $789 billion in R&D in 2021, up $72 billion from 2020. On the other hand, Switzerland reported a total spending of only $35 billion in 2022. These factors combined, position the United States as the **most innovative country** in the world.

**US R&D spending outpaces China's.*

Matthews 24 — David Matthews, international editor of Science|Business, holds a BA in Modern History from the University of Oxford, 2024 ("US holds off China challenge in global R&D spending race," Science|Business, March 14th, Available Online at https://sciencebusiness.net/news/international-news/us-holds-china-challenge-global-rd-spending-race, Accessed 08-18-2024)

Closely watched science and technology indicators from Washington DC show the **US** is still the world's **biggest spender** on research and development, confounding expectations that China would takeover the lead.

A **faltering** Chinese **economy** and plummeting **birthrates** have driven a conversation over whether the world has reached '**peak China'**, where Beijing's economic and geopolitical ascendancy over the US no longer looks **inevitable**. Now, a similar debate is happening in the domain of R&D.

On 13 March the US's National Science Board (NSB), which advises Congress and the president, released its biennial State of US Science & Engineering report, a gold-standard set of data on the shape of the world's R&D effort.

It shows that the US still retains a **healthy lead**, spending **$806 billion** on R&D, both public and private, in 2021, the latest year the report calculates. China spent **$668 billion**.

While China has been catching the US over the last two decades, on the basis of these statistics, it **no longer** looks on a **trajectory** to **overtake** it.

"We've all been waiting to see when those two curves cross, and they haven't crossed," said Arati Prabhakar, director of the White House Office of Science and Technology Policy at an event launching the report.

Back in 2018, it looked inevitable that China would soon take the top spot. In that year, the NSB issued a warning that on current trends, it expected China to surpass the US in R&D investment by the end of the year.

But in recent years the US has seen a "**huge surge**" in corporate spending on R&D, particularly by IT and pharmaceutical companies, Prabhakar said.

"It came from **businesses**," she said. "What you see in these numbers is the **intensification** of our innovation economy."

The statistics bear this out: from 2018 to 2021, business R&D spending **ballooned** by more than a third, to over $600 billion. Meanwhile, federally funded R&D stayed **roughly flat** as a proportion of the economy, Prabhakar noted.

Private sector R&D is now so **important** in the US than it funds almost as great a proportion of basic research (36%) as the federal government (40%).

6 No Taiwan Invasion—2NC/1NR

Extend China won't invade Taiwan —

A) Cost — it's expensive and takes too long to get soldiers across the water.

B) Detection — it's impossible for them to be offensive without Taiwan knowing.

C) Strategic Disadvantage — Taiwan can take out Chinese ships in port, and their knowledge of the land makes it impossible for China to have a successful invasion

That's all Sacks

D) MAD & Structural Problems Check *— China can't fight the US without destroying themselves, and political and economic restraints block invasion.*

Swaine 21 — Michael D. Swaine, Director of the East Asia Program at the Quincy Institute for Responsible Statecraft, Senior Associate at Carnegie, PhD in Political Science from Harvard University, BA in Chinese Studies from The George Washington University, 2021("China Doesn't Pose an Existential Threat for America," *Foreign Policy*, April 21st, Available Online at https://foreignpolicy.com/2021/04/21/china-existential-threat-america/, Accessed 08-06-2024)

In fact, there isn't much actual evidence to support the notion of China as an existential threat. That does not mean that China is not a threat in some areas, but Washington needs to approach this issue based on the facts, not dangerous rhetoric. Unfortunately, right-sizing the challenges that China poses seems to be an impossible task for Washington. In the most basic, literal sense, an existential threat means a threat to the physical existence of the nation through the possession of an ability and intent to exterminate the U.S. population, presumably via the use of highly lethal nuclear, chemical, or biological weapons. A less conventional understanding of the term posits the radical erosion or ending of U.S. prosperity and freedoms through economic, political, ideational, and military pressure, thereby in essence destroying the basis for the American way of life. Any threats that fall below these two definitions do not convey what is meant by the word "existential." As a military power, China has no ability to destroy the United States without destroying itself. China's nuclear capabilities are far below those of the United States, and its conventional military, while regionally potentially powerful, has a fraction of the budget of that of the United States. Some argue that China could militarily push the United States out of Asia and dominate that region, denying the country air and naval access and hence support for critical allies. This would presumably have an existential impact by virtue of the supposedly critical importance of that region to the stability and prosperity of the United States. Yet there are no signs that Washington is losing either the will or the capacity to remain a major military actor in the region and one closely connected to major Asian allies, which are themselves opposed to China dominating the region. In reality, the greater danger in Asia is that Washington could so militarize its response to China that its actions and policies become repugnant even to U.S. allies. This leaves the unconventional threats. Here they are presumably twofold: economic and technological, and in the realm of ideas and influence operations within the United States and other Western countries, including the export of

China's so-called "model" of authoritarian rule to the rest of the world. The former threats would presumably consist of China attaining a level of total superiority over both economic and technological levers of influence globally and with regard to the United States (perhaps combined with a successful military blocking of U.S. sea lines of communication) that would so impoverish the country as to threaten its existence as a stable and prosperous democracy and bring it under Chinese control. Presumably, the specific basis of such leverage would consist of near-absolute global Chinese dominance over both trade and investment relations and supply chains with the United States and other countries and over all the key technologies driving future growth and military capabilities. It is virtually inconceivable that China could achieve such a level of dominance over the United States. The United States possesses abundant energy, human, technological, and other resources; a huge and dynamic domestic market; enormous levels of accumulated wealth and capital stocks; and the globe's financial reserve currency. In contrast, while China boasts a highly entrepreneurial and dynamic workforce, it labors under major structural and political constraints such as insufficient arable land, a rapidly aging society, a heavy reliance on energy imports, and stifling ideological and state-centered controls across society. Beijing has certainly used its economic leverage (such as market access) to pressure foreign companies and governments to support Chinese policies or stop what it regards as unacceptable behavior, e.g., regarding Taiwan. While such economic coercion is by no means unique to China, it certainly can erode freedom of speech, thus threatening one of democracy's core principles. But this hardly rises to the level of an existential threat to American values, given both the limited reach of Chinese economic power and the countervailing global economic power and political influence of democratic states. Some observers claim that Beijing could somehow set standards in critical technology areas and install tech hardware around the world, to the extent that China would be able to relegate the United States to a permanently inferior status in both the commercial and military realms, thus threatening the very existence of the country. This is also highly unlikely. Chinese companies are certainly participating in standard-setting in key areas, including 5G. But this process is highly competitive globally, and U.S., Asian, and European companies all hold major portions of the standards and the standard-essential patents that undergird the global technology ecosystem. There is little if any chance that Chinese companies could come to dominate this process. Many tech experts state that the most likely worst-case outcome of Chinese gains regarding standards and hardware